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AI-Powered Real-Time Exam Invigilation Support System Using YOLO and Byte Track

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***Abstract*— The problem of cheating during examinations is a major threat to the integrity of examination processes. The purpose of our project is to provide a real-time system of Exam Invigilation using computer vision techniques to improve the efficiency of the examination process. The object detection algorithm used is YOLO (You Only Look Once), which enables the system to detect the students, invigilators, as well as other suspicious objects such as the use of a mobile phone or the use of notes within the examination hall. In addition, the system uses the Byte Track algorithm to perform the object tracking process. The outcome of the project is to provide recorded video or real-time video as input to the system, which is then used to detect cases of cheating during the examination process. The use of the IP WebApp enables the system to capture the activities of students in real-time using the camera of a mobile phone, with the images then being viewed on the PC screen with red boxes indicating the suspicious students and green boxes indicating the normal students. The proposed solution processes the data in real-time with minimal human involvement, which improves the level of scalability, accuracy, and fairness of the exam invigilation process.**

Keywords*— *AI-Powered Invigilation, YOLO, Byte Track, Object Detection, Multi-Object Tracking, Suspicious Behavior Monitoring, Quality Education (SDG-4), Industry Innovation Infrastructure (SDG-9), Peace justice and Strong Institution (SDG-16).

I. INTRODUCTION

Examinations are vital to assess the knowledge and progress of a student. Nevertheless, the increasing rate of malpractices during examinations has turned out to be a major threat to academic integrity. Human invigilators can only monitor a few students at a time. In a large examination hall or an online exam with hundreds of students, it is almost impossible to monitor all malpractices (such as copying from other students, using mobile phones, and whispering answers to students). That is why large-scale malpractices during examinations go undetected using traditional methods. There is an increasing need to develop a more reliable system to ensure fair examination practices using technology. Our project attempts to address this problem by developing an automated surveillance system to monitor examination halls. It is designed to intelligently monitor examination halls and detect malpractices during examinations.
students, detect suspicious activities, and reduce the chances of cheating.
In the proposed system, we are using a powerful object detection technique named YOLO, which is efficient and accurate for object detection. The proposed system can use the YOLO technique to detect students, invigilators, as well as forbidden objects like mobile phones and cheat sheets. In order to improve the efficiency of the proposed system for monitoring activities and detecting cheating attempts, we plan to use a powerful technique named Byte Track, which is efficient for tracking multiple objects. The technique can be used to track students and objects to detect suspicious activities.
The rest of the paper is presented as follows: In Section II, a brief overview of the existing literature regarding the existing approaches to the usage of AI in the examination monitoring system is presented .In Section III, the methodology is presented. In this section, a brief description of data acquisition, input data pre-processing, and the real-time monitoring system using the YOLO algorithm and the Byte Track algorithm is presented. In Section IV, a brief description of the evaluation of the proposed methodology is presented. In Section V, a brief description of the conclusion of the paper is presented. In this section, a brief description of the major contributions, future scope, and scope of improvement in the automated examination invigilation system is presented.

II. RELATED WORK

Arumugam [1] proposed a cheating detection model based on deep learning techniques and ensemble techniques such as YOLO, faster R-CNN, and multimodal fusion techniques. It was reported to have a high accuracy compared to other traditional models. However, it was challenged by the computational overhead. Rajasekar and Vijayarani [2]

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proposed a cheating detection system based on a deep learning technique and computer vision techniques. It was reported to have promising results. However, it was challenged by the issue of crowded rooms. Nagajothi et al. [3] proposed a prototype system based on artificial intelligence to detect cheating in exams based on the use of the YOLO algorithm and facial recognition techniques. It was challenged by the issue of a single camera and behavioral studies. Ghita et al. [4] proposed a review of various studies based on 58 studies published between 2010 and 2021 to detect cheating in online exams. It was challenged by the issue of traditional techniques.

The idea of a system for online proctoring was proposed by Kumar et al. [5]. using AI technology and deep learning techniques along with computer vision and multimodal features. However, false positives were identified in the system. An automated system for online exam proctoring was suggested by Patel and Mehta [6] using deep learning and early warning techniques. However, the system's performance was affected by environmental noise. A smart proctoring system was suggested by Kaur et al. [7] using deep learning techniques for academic integrity. However, the system used an improved version of YOLOv8 and required fine-tuning for various scenarios. The techniques of YOLO and LSTM were used by Reddy et al. [8] for real-time monitoring in classrooms. Improvements were identified by the suggested system; however, the system's performance was affected by generalization issues. A system was suggested by Suardana et al. [9] using YOLOv5 techniques for detecting cheating movements in offline examinations.

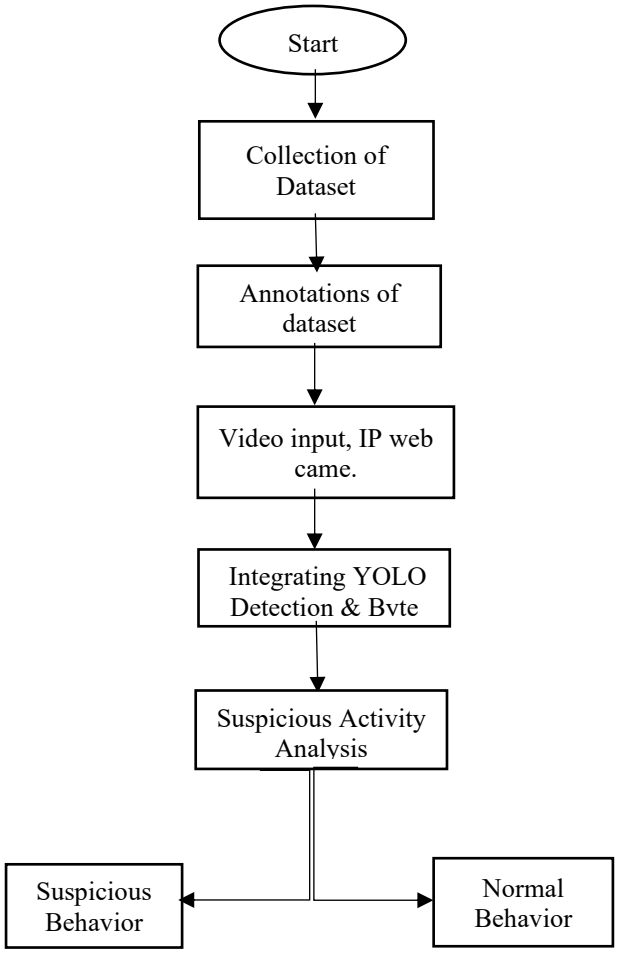
In a study by Patel and Singh, who used YOLOv8 to detect communication and material usage inside a closed room during exams, the results showed a high adaptability. However, there was a decline in crowded rooms. Ahmed and Rao proposed a system that used face detection using YOLO, tracking of gaze, and audio recognition. The system was effective in detecting all forms of cheating in online exams. However, it requires a high computational power. Banerjee and Mehta used pose estimation and object detection using YOLO to detect signal usage. However, there was a decline in accuracy due to occlusions. Zhang and Huang used YOLOv8 with attention to detect gestures. It was effective in detecting cheating gestures. However, it requires optimization of the dataset. Holi et al. proposed a system using ResNet50 to classify six forms of cheating behaviors. The system was 96.57% accurate and used a small dataset. Radwan et al. proposed a system using Graph Convolutional Networks to detect cheating behaviors AUC of 87.58% while tackling localization issues, although limited in the incorporation of multimodal cues such as gaze and background.

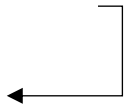
The majority of the studies conducted between 2022 and 2025 on the detection of malpractice in exams employed computer vision techniques such as YOLO and some studies employed CNN or RCNN for the same. Some studies employed enhanced detection systems using eye movement tracking for the detection of abnormal gaze patterns and hence increased the accuracy of cheating detection systems. These systems can be rated as good because they can detect cheating easily and accurately and can function using normal cameras. However, the systems have several issues such as the possibility of the student hiding from the camera, poor lighting, poor cameras, small training sets, and privacy issues.

Our project utilizes YOLO with Byte Track to ensure accurate and precise detection and tracking, even if students change their location or hide. It is also suitable for low-light conditions and can function with normal cameras, eliminating any issues due to low visibility and low-processing capabilities. Additionally, using a large and balanced dataset will help us avoid the shortcomings of a small and imbalanced dataset. Moreover, our system will be more reliable due to the addition of features such as multiple cameras and eye tracking.

III. METHODOLOGY

A. WorkFlow





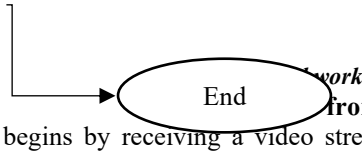
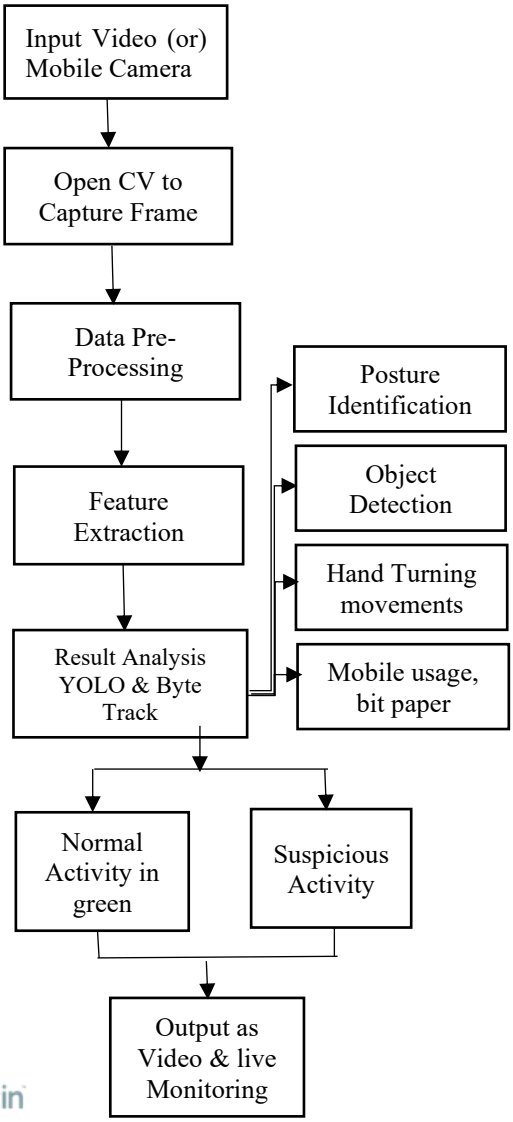


Fig:1 flowchart

Collection of Dataset: Collect images using exam environments.Vision Input using IP Webcam: In the proposed approach, video input is obtained through the IP Webcam application, which streams live camera footage to the system for further processing and analysis. The proposed system uses the YOLO model to identify objects in video. After that, the ByteTrack model is used to consistently track these objects as they move from one frame to the next. Analysis of Suspicious Activities: In the proposed system, various activities such as unusual hand gestures, the use of mobile phones, passing small pieces of paper, or signaling to others are analyzed to check if the activity is suspicious or not.

Classification of Behavior: If the activity score is more than the threshold, the activity is classified as suspicious. Otherwise, the activity is classified as normal, and the monitoring continues without any interruptions.

B. Proposed Methodology



work

from Camera: The system begins by receiving a video stream from a mobile phone camera or any available video recording device. This video serves as the primary input for the monitoring process.

Feature Extraction: These extracted features help the system understand the posture and movement of objects in the scene

Frame Capture using OpenCV: The video is processed by using OpenCV, and the video is split into multiple frames. Each frame is considered an image so that the activities can be observed step by step.

Data Preprocessing: The captured frames are further processed to improve the quality of the images and eliminate unwanted noise. The process may include the following steps: Images may be converted to grayscale images. The images may be resized. The images may be smoothened. The foreground may be separated from the background.

Posture and Object Identification: The system is capable of recognizing the postures of humans, such as sitting, standing, or bending. The system is also capable of recognizing objects like mobile phones or pieces of paper.

C. Algoritham

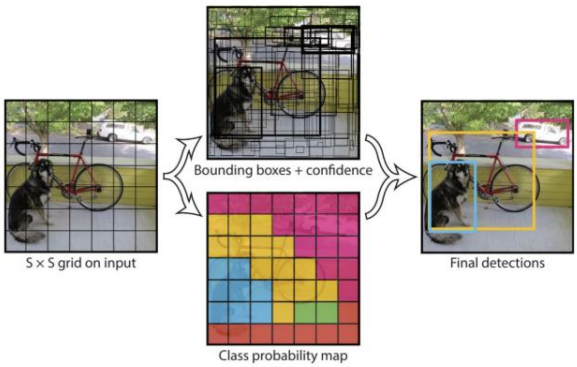


Fig:3 YOLO work

YOLO is a deep learning algorithm based on the principle of dividing the given image into a grid, predicting the bounding box coordinates, and the probabilities of the objects present in the individual cells of the grid. It is a very fast algorithm.

YOLO (You Only Look Once): This algorithm can be used to detect objects in real-time in the exam hall.

Byte Track: This algorithm can be used to track objects in the video.

Byte Track is an extension of SORT, which stands for Simple Online Real-Time Tracking. It tracks objects by linking high-confidence and low-confidence detection results, thus preventing the objects from getting lost even if the confidence level falls.

Each student is uniquely identified in the frame. It ensures the continuous monitoring of the students' behavior.

Each student's position is tracked in real-time to identify suspicious activity.

IV. RESULT ANALYSIS AND DISSCUSSION

In this section, the results that were obtained after training the YOLOv8 model using the custom dataset will be discussed. The model was subjected to testing in order to determine the extent to which the model can identify various activities of the students, such as normal activities, suspicious activities, and the activities of invigilators in the classroom. In order to determine the performance of the model, various measures were investigated, including the precision, recall, F1 score, and the average average precision of the model. Additionally, the confusion matrix and training loss were utilized in determining the performance of the model.

A. F1-Confidence Curve

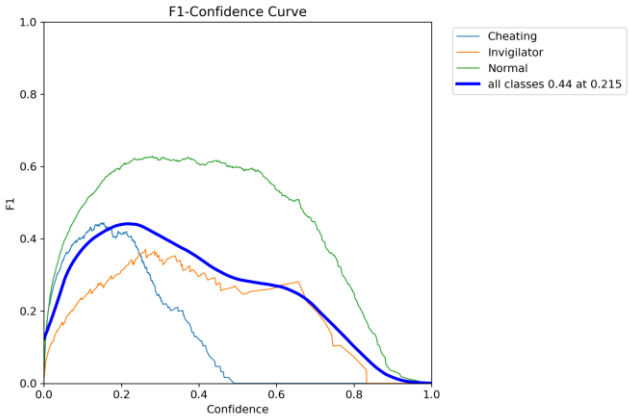


Fig:4 F1-Confidence curve

The F1-Confidence curve shows the effect of the confidence level of the model on the precision and recall of the model. From the results obtained in Figure 4, the Normal category has the highest F1-score compared to the Cheating and Invigilator categories. This shows that the model is more reliable in recognizing students with regular behavior compared to the others.

B. Precision-Recall Curve

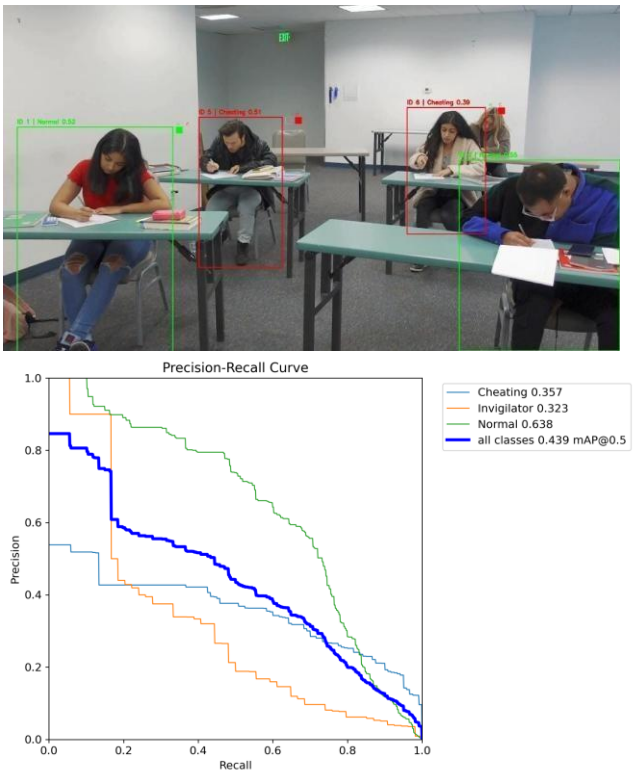


Fig:5 Precision-Recall Curve

The Precision-Recall curve in Figure 5 illustrates the trade-off between precision and recall. The 'Normal' class has again recorded the best results with a precision-recall of 0.638. The 'Cheating' class has recorded a precision-recall of 0.357, while the 'Invigilator' class has recorded 0.323.The mAP@0.5 for all the classes is 0.439. This is an average level of detection capability.

C. Confusion Matrix

The confusion matrix gives a clear view of the correct or incorrect classification. It shows the model incorrectly classifies the 'Cheating' students as 'Normal'. This results in the low precision and recall values for the 'Cheating' class. Also, the model incorrectly classifies the 'Invigilator' class as 'Normal'.

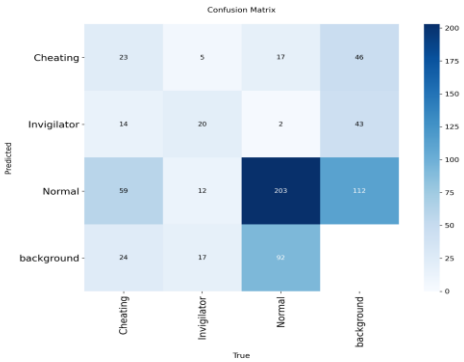


Fig:6 Confusion matrix

D. Output

Fig:7 Suspicious and normal

In Figure 7 Images, we present an overview of how the proposed invigilation system identifies and classifies suspicious behaviors during examinations.
In one of the images, a student is observed to repeatedly turn towards a neighboring desk, which is classified by the system as "Cheating".

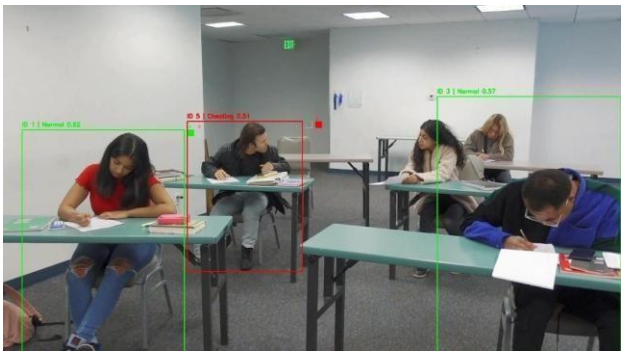


Fig: 8 Suspicious and normal

The figure 8 illustrates two students who are in close proximity to one another, which implies that they are exchanging something. This is considered an unusual act by the system. The visual results obtained above have confirmed that with the incorporation of YOLO for object detection and Byte Track for tracking, it is possible to identify malpractice correctly. The bounding boxes and labels used in the images are not only helpful for the process of monitoring but also act as evidence to justify the purpose of maintaining integrity with minimum human involvement.

V. CONCLUSION

Our project can be seen as a tireless and highly observant invigilator that has been trained using the most advanced forms of AI algorithms such as YOLO and Byte Track for ensuring fair and secure examination centers. It can identify each and every student and their activities while at the same time spotting any abnormal or suspicious behavior such as looking at another person's paper or passing notes. This system can be distinguished from human invigilators in that, unlike them, this system can keep track of all the students at the same time without any form of distraction or difficulty. This smart invigilation system can be highly beneficial in ensuring that honest students do not suffer at any disadvantage in schools, colleges, or in the case of online tests and exams in large centers.

VI. FUTURE WORK

The future of such a system would allow for improvements to be made, such as incorporating an audio-based monitoring feature. This would enable it to detect whispering, talking, or even the possible use of hidden electronic devices that might not be easily detected via camera-based observation. Another feature that would greatly enhance its functionality is incorporating a face recognition feature. This would automatically identify students, hence recording attendance. This ensures that only registered candidates are inside the room during an examination. If we were to implement a system for an open-book examination, it would also allow students to refer to books or any notes while taking an examination. At the same time, it would continue to monitor students' behavior to ensure that they are not sharing information with others or using restricted devices.

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